

Endpoint Forensics and Threat Hunting Lab Using Velociraptor

Project Overview

This project demonstrates the deployment and use of Velociraptor for endpoint visibility, digital forensics, and threat hunting. A Windows 11 endpoint was connected to a Velociraptor server hosted on Ubuntu Linux to collect and analyze endpoint telemetry.

The objective was to establish a baseline of normal system activity and identify potential indicators of compromise (IOCs) through process, network, service, and persistence analysis.

Environment

Server

- Ubuntu Linux
- Velociraptor Server v0.77.1
- IP Address: 192.168.X.XXX

Endpoint

- Microsoft Windows 11 Home
- Hostname: Edward
- Velociraptor Agent v0.77.1
- IP Address: 192.168.X.XXX

Additional Security Tools

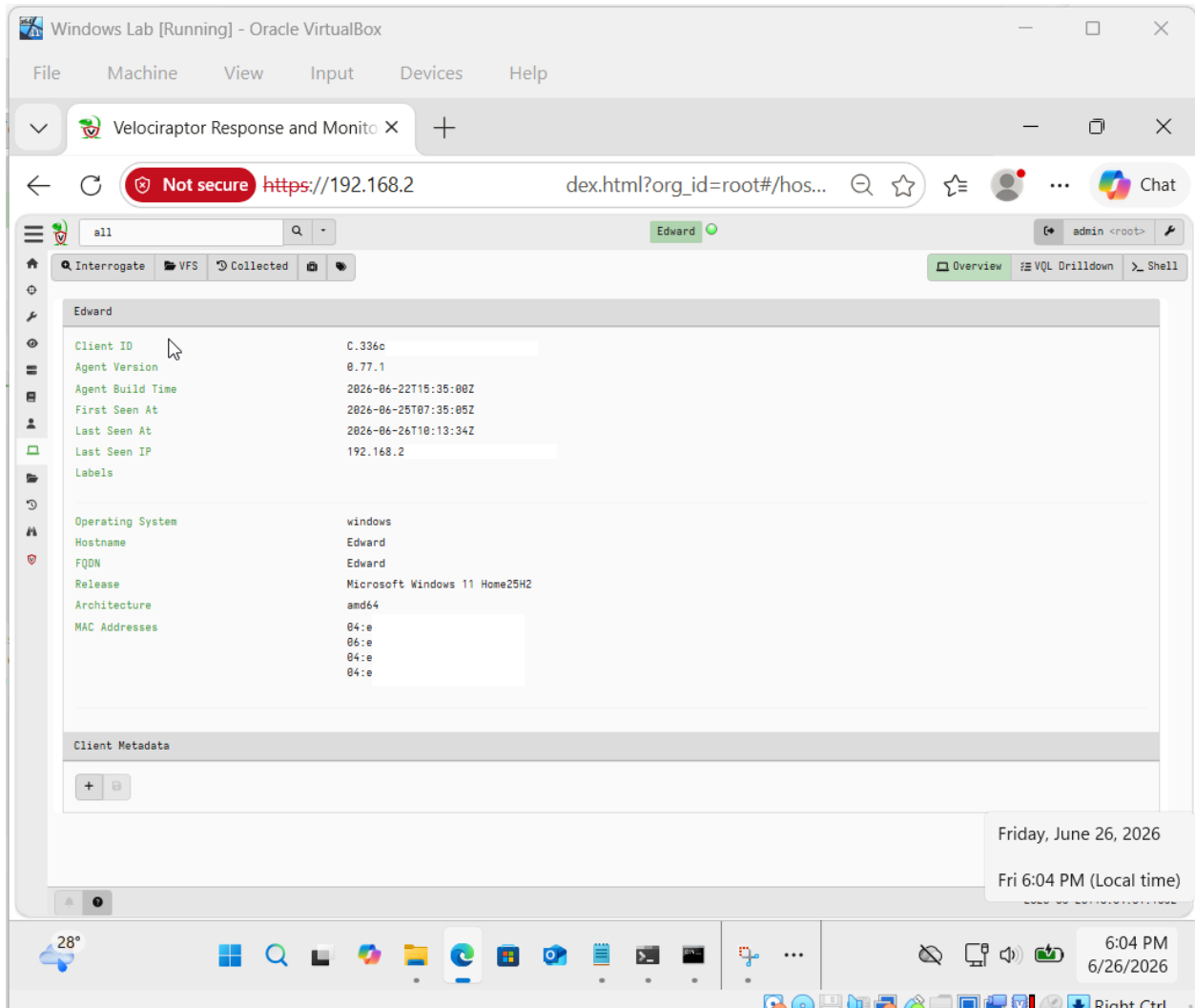
- Wazuh SIEM
- Suricata IDS

Objectives

1. Deploy and configure a Velociraptor server.
2. Connect a Windows endpoint to the server.
3. Collect forensic artifacts from the endpoint.
4. Establish a baseline of normal system activity.
5. Perform basic threat hunting and anomaly detection.

Methodology

1. Client Enrollment



The Windows endpoint was successfully enrolled into Velociraptor and verified through the client information page.

Collected information included:

- Hostname
- Operating System Version
- Agent Version
- Network Information
- Hardware Architecture

2. Process Analysis

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.DBUEACNB2PC8I	Windows.System.Pslist	2026-06-25T08:22:10.117Z	2026-06-25T08:22:20.387Z	admin	0 b	290
✓	F.D8UDKAAPIC1H2	Generic.Client.Info	2026-06-25T07:35:05.630Z	2026-06-25T07:35:05.697Z	InterrogationService	0 b	26

CreateTime	Pid	Ppid	TokenIsElevated	Name	CommandLine	Exe	TokenInfo	Hash	AuthenticCode	Username	WorkingSetSize
2026-06-18T02:48:32.074Z	4	0	true	System						NT AUTHORITY\SYSTEM	147456
2026-06-18T02:48:28.528Z	176	4	false	Registry						NT AUTHORITY\SYSTEM	23572480
2026-06-18T02:48:32.078Z	632	4	true	smss.exe	\SystemRoot\System32\smss.exe	C:\Windows\System32\smss.exe		{ "MD5": "d997898238c9ffde81d6d3aef879e1", "SHA1": "bd934d7c1885852331a"	{ "Filename": "C:\\Windows\\System32\\smss.exe", "ProgramName": "Microsoft Windows", "PublisherLink": null, "MoreInfoLink": null }	NT AUTHORITY\SYSTEM	167936

Artifact Used:

- Windows.System.Pslist

Purpose:

- Enumerate all running processes.
- Identify suspicious executables.
- Detect process masquerading attempts.

Findings:

- 290 active processes were identified.
- Core Windows processes such as explorer.exe, svchost.exe, lsass.exe, winlogon.exe, and services.exe were present.
- No suspicious process names or unusual executable paths were identified.

Assessment:

- No evidence of malicious process execution was observed.

3. Network Analysis

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.D8UEDD035JQCE	Windows.Network.Netstat	2026-06-25T08:28:36.020Z	2026-06-25T08:28:36.705Z	admin	0 b	230
✓	F.D8UEACN82PC8I	Windows.System.Pslist	2026-06-25T08:22:10.117Z	2026-06-25T08:22:20.387Z	admin	0 b	290
✓	F.D8UDKAAPICIH2	Generic.Client.Info	2026-06-25T07:35:05.630Z	2026-06-25T07:35:05.697Z	InterrogationService	0 b	26

Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port	Timestamp
1420	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	135	0.0.0.0	0	2026-06-18T02:48:37Z
4	System	IPv4	TCP	LISTEN	192.	139	0.0.0.0	0	2026-06-25T06:21:14Z
8784	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	5040	0.0.0.0	0	2026-06-25T07:01:44Z
10216	Spotify.exe	IPv4	TCP	LISTEN	127.0.0.1	7768	0.0.0.0	0	2026-06-24T07:49:25Z
4048	AsusSoftwareManager.exe	IPv4	TCP	LISTEN	127.0.0.1	24830	0.0.0.0	0	2026-06-18T02:48:43Z
2636	AppleMobileDeviceService.exe	IPv4	TCP	LISTEN	127.0.0.1	27015	0.0.0.0	0	2026-06-18T02:48:40Z
4	System	IPv4	TCP	LISTEN	127.0.0.1	28385	0.0.0.0	0	2026-06-18T02:48:44Z
4	System	IPv4	TCP	LISTEN	127.0.0.1	28390	0.0.0.0	0	2026-06-18T02:48:44Z
24436	chrome.exe	IPv4	TCP	ESTAB	192.1	49200	74.125	443	2026-06-25T08:25:41Z
24436	chrome.exe	IPv4	TCP	ESTAB	192.1	49209	57.144	443	2026-06-25T07:16:17Z

Artifact Used:

- Windows.Network.Netstat

Purpose:

- Identify active network connections.
- Detect unauthorized outbound communication.
- Review listening services.

Findings:

- Network activity was consistent with normal operating system and user activity.
- No suspicious remote connections were identified.

Assessment:

- No indicators of command-and-control (C2) communications were observed.

4. Service Analysis

The screenshot shows a web application interface. At the top, there's a browser address bar with 'https://192.168.1.100/results'. Below it, a navigation bar includes a search bar with 'all', a user profile 'Edward' with a 'Connected' status, and a dropdown menu showing 'admin <root>'. The main content area has a tabbed interface with 'Artifact Collection', 'Uploaded Files', 'Requests', 'Results' (selected), 'Log', and 'Notebook'. The 'Results' tab displays a table of artifacts:

State	FlowId	Artifacts	Created	Last Active	Creator	Size	Rows
✓	F.DBUEFGN22624	Windows.System.Services	2026-06-25T08:33:06.504Z	2026-06-25T08:33:08.557Z	admin	0 b	298
✓	F.D8UEDD035JQCE	Windows.Network.Netstat	2026-06-25T08:28:36.820Z	2026-06-25T08:28:36.705Z	admin	0 b	238
✓	F.D8UEACN82PC8I	Windows.System.Pslist	2026-06-25T08:22:10.117Z	2026-06-25T08:22:20.387Z	admin	0 b	298
✓	F.D8UDKAAPIC1H2	Generic.Client.Info	2026-06-25T07:35:05.630Z	2026-06-25T07:35:05.697Z	InterrogationService	0 b	26

Below the artifacts table, there's a section titled 'Windows.System.Services' with a sub-table showing details for two services:

State	Name	DisplayName	Status	Pid	ExitCode	StartMode	PathName	ServiceType	UserAccount	Created	ServiceDll
Stopped	ADPSvc	Aggregated Data Platform Service	OK	0	1077	Manual	C:\WINDOWS\system32\svchost.exe -k LocalServiceAndNoImpersonation -p	Share Process	NT AUTHORITY\LocalService	2026-04-11T09:46:12.990Z	C:\WINDOWS\system32\adpsvc.dll
Stopped	ALG	Application Layer Gateway Service	OK	0	1077	Manual	C:\WINDOWS\system32\alg.exe	Own Process	NT AUTHORITY\LocalService	2024-04-01T07:26:32.631Z	

Artifact Used:

- Windows.System.Services

Purpose:

- Enumerate running Windows services.
- Identify unauthorized or suspicious services.

Findings:

- Running services corresponded to standard Windows functionality.
- Service executable paths appeared legitimate.

Assessment:

- No unauthorized services were detected.

5. Persistence Analysis

The screenshot shows a web application interface with a browser address bar displaying "https://192.1". The application has a top navigation bar with a search bar containing "all", a status indicator "Edward Connected", and a user dropdown "admin <root>". Below the navigation bar is a table with columns: State, FlowId, Artifacts, Created, Last Active, Creator, Mb, and Rows. The table contains four rows of data, all with a "State" of "✓". The first row is highlighted in green.

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.D8UEGD1NV3IHA	Windows.System.TaskScheduler	2026-06-25T08:35:00.035Z	2026-06-25T08:35:06.228Z	admin	0 b	282
✓	F.D8UEFGG22624	Windows.System.Services	2026-06-25T08:33:06.504Z	2026-06-25T08:33:08.557Z	admin	0 b	298
✓	F.D8UEDD0J5JQCE	Windows.Network.Netstat	2026-06-25T08:28:36.020Z	2026-06-25T08:28:36.705Z	admin	0 b	230
✓	F.D8UEACNB2PC8I	Windows.System.Pslist	2026-06-25T08:22:10.117Z	2026-06-25T08:22:20.387Z	admin	0 b	290

Below the table is a tabbed interface with tabs: Artifact Collection, Uploaded Files, Requests, Results, Log, and Notebook. The "Results" tab is selected, showing a detailed view of the first artifact. The view includes a search bar with "Windows.System.TaskScheduler/Analysis" and a table with columns: TaskName, Mtime, Command, Arguments, UserId, RunLevel, LogonType, StateChange, ComHandler, RegistrationTime, and StartBoundary.

TaskName	Mtime	Command	Arguments	UserId	RunLevel	LogonType	StateChange	ComHandler	RegistrationTime	StartBoundary
\ASUS Update Checker 2.0	2026-04-10T18:00:56.508Z	c:\windows\system32\driverstore\files\rep\ository\asus\ci.inf_ amd64_4bff547a9eb4d ecc\asussoftw aremanager\as usupdatecheck er.exe		S-1-5-18	HighestAvailab le					

Artifact Used:

- Windows.System.TaskScheduler

Purpose:

- Identify persistence mechanisms commonly used by attackers.
- Review scheduled tasks for unauthorized activity.

Findings:

- Scheduled tasks were associated with legitimate Windows maintenance and update operations.
- No suspicious persistence mechanisms were observed.

Assessment:

- No malicious persistence techniques were identified.

Threat Hunting Observations

The endpoint was examined for common indicators of compromise including:

- Suspicious process names
- Processes executing from temporary directories
- Unauthorized network connections
- Unknown Windows services
- Malicious scheduled tasks

No indicators of compromise were identified during the baseline assessment.

Skills Demonstrated

- Endpoint Forensics
- Digital Forensics and Incident Response (DFIR)
- Threat Hunting
- Windows Artifact Analysis
- Endpoint Visibility
- Security Monitoring
- Velociraptor Administration
- Security Operations Center (SOC) Operations

Results

The Windows 11 endpoint was successfully investigated using Velociraptor. Multiple forensic artifacts were collected and analyzed to establish a baseline of normal activity.

The investigation found no evidence of malicious processes, unauthorized services, suspicious network connections, or persistence mechanisms. This baseline can be used for future comparisons during incident response and threat hunting exercises.

Conclusion

This project demonstrates the practical use of Velociraptor as an endpoint visibility and forensic investigation platform. Through the collection and analysis of endpoint artifacts, a baseline of system activity was established, providing a foundation for future threat hunting and incident response activities.